

SKILL – 1

GIT VERSION CONTROL

STEP 1 : Create a new project folder, initialize Git using git init, and configure Git with your username and email.

1. Open Git Bash or Terminal

1. Create a new folder

```
MINGW64:/c/Users/HP/git-skill1  
  
HP@LAPTOP-PKGHGOCV MINGW64 ~  
$ mkdir git-skill1  
  
HP@LAPTOP-PKGHGOCV MINGW64 ~  
$ cd git-skill1
```

2. Initialize Git

```
HP@LAPTOP-PKGHGOCV MINGW64 ~/git-skill1  
$ git init  
Initialized empty Git repository in C:/Users/HP/git-skill1/.git/
```

3. Configure username and email

```
HP@LAPTOP-PKGHGOCV MINGW64 ~/git-skill1 (master)  
$ git config --global user.name "ChikkaTejaswi"  
  
HP@LAPTOP-PKGHGOCV MINGW64 ~/git-skill1 (master)  
$ git config --global user.email "2400032263@kluniversity.in"
```

4. Verify configuration

```
HP@LAPTOP-PKGHGOCV MINGW64 ~/git-skill1 (master)
$ git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=schannel
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
user.name=ChikkaTejaswi
user.email=2400032263@kluniversity.in
core.repositoryformatversion=0
core.filemode=false
core.bare=false
core.logallrefupdates=true
core.symlinks=false
core.ignorecase=true
```

STEP 2:

1. Create two files

```
HP@LAPTOP-PKGHGOCV MINGW64 ~/git-skill1 (master)
$ touch main.java notes.txt
```

2. Add sample content

```
HP@LAPTOP-PKGHGOCV MINGW64 ~/git-skill1 (master)
$ echo "Hello Git" > main.java

HP@LAPTOP-PKGHGOCV MINGW64 ~/git-skill1 (master)
$ echo "Git skill 1 notes" > notes.txt
>
> exit
> ^C

HP@LAPTOP-PKGHGOCV MINGW64 ~/git-skill1 (master)
$ echo "Git skill 1 notes" > notes.txt
```

3. Check repository status

```
HP@LAPTOP-PKGHGOCV MINGW64 ~/git-skill1 (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    main.java
    notes.txt

nothing added to commit but untracked files present (use "git add" to track)
```

4. Add files to staging

```
HP@LAPTOP-PKGHGOCV MINGW64 ~/git-skill1 (master)
$ git add .
warning: in the working copy of 'main.java', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'notes.txt', LF will be replaced by CRLF the next time Git touches it
```

5. Check status again

```
HP@LAPTOP-PKGHGOCV MINGW64 ~/git-skill1 (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   main.java
    new file:   notes.txt
```

✓ Files

will appear in green (staged).

STEP 3

```
HP@LAPTOP-PKGHGOCV MINGW64 ~/git-skill1 (master)
$ git commit -m "Initial commit with main.java notes.txt"
[master (root-commit) 2e8e9e9] Initial commit with main.java notes.txt
2 files changed, 2 insertions(+)
create mode 100644 main.java
create mode 100644 notes.txt
```

STEP 4:

3. Connect local repo to GitHub

```
HP@LAPTOP-PKGHGOCV MINGW64 ~/git-skill1 (master)
$ git remote add origin https://github.com/ChikkaTejaswi/git-skill1
```

4. Push code

```
HP@LAPTOP-PKGHGOCV MINGW64 ~/git-skill1 (master)
$ git branch -M main

HP@LAPTOP-PKGHGOCV MINGW64 ~/git-skill1 (main)
$ git push -u origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (4/4), 315 bytes | 157.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/ChikkaTejaswi/git-skill1
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```



Code is now visible on GitHub.

STEP 5: Create and switch branch

```
HP@LAPTOP-PKGHGOCV MINGW64 ~/git-skill1 (main)
$ git checkout -b feature-update
Switched to a new branch 'feature-update'
```

1. Modify file

```
HP@LAPTOP-PKGHGOCV MINGW64 ~/git-skill1 (feature-update)
$ echo "Feature update added" >> main.java
```

2. Add and commit

```
HP@LAPTOP-PKGHGOCV MINGW64 ~/git-skill1 (feature-update)
$ git add main.java
warning: in the working copy of 'main.java', LF will be replaced by CRLF the next time Git touches it

HP@LAPTOP-PKGHGOCV MINGW64 ~/git-skill1 (feature-update)
$ git commit -m "Added feature update"
[feature-update 9f316d7] Added feature update
1 file changed, 1 insertion(+)
```

STEP 6:

1. Switch to main

```
HP@LAPTOP-PKGHGOCV MINGW64 ~/git-skill1 (feature-update)
$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
```

2. Create new branch

```
HP@LAPTOP-PKGHGOCV MINGW64 ~/git-skill1 (main)
$ git checkout -b bug-fix
Switched to a new branch 'bug-fix'
```

3. Modify file

```
HP@LAPTOP-PKGHGOCV MINGW64 ~/git-skill1 (bug-fix)
$ echo "Bug fixed here" >> main.java
```

4. Add and commit

```
HP@LAPTOP-PKGHGOCV MINGW64 ~/git-skill1 (bug-fix)
$ git add main.java
warning: in the working copy of 'main.java', LF will be replaced by CRLF the next time Git touches it

HP@LAPTOP-PKGHGOCV MINGW64 ~/git-skill1 (bug-fix)
$ git commit -m "Big fix applied"
[bug-fix ae6a41b] Big fix applied
1 file changed, 1 insertion(+)
```

STEP 7: Switch to main

```
HP@LAPTOP-PKGHGOCV MINGW64 ~/git-skill1 (bug-fix)
$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
```

1. Merge feature branch

```
HP@LAPTOP-PKGHGOCV MINGW64 ~/git-skill1 (main)
$ git merge feature-update
Updating 2e8e9e9..9f316d7
Fast-forward
 main.java | 1 +
1 file changed, 1 insertion(+)
```

2. Merge bug-fix branch

```
HP@LAPTOP-PKGHGOCV MINGW64 ~/git-skill1 (main)
$ git merge bug-fix
Auto-merging main.java
CONFLICT (content): Merge conflict in main.java
Automatic merge failed; fix conflicts and then commit the result.
```

STEP 8: Add resolved file

```
HP@LAPTOP-PKGHGOCV MINGW64 ~/git-skill1 (main|MERGING)
$ git add main.java
```

1. Commit resolution

```
HP@LAPTOP-PKGHGOCV MINGW64 ~/git-skill1 (main|MERGING)
$ git commit -m "Resolved merge conflict"
[main 8dd0d4a] Resolved merge conflict
```

✅ FINAL CHECK

```
HP@LAPTOP-PKGHGOCV MINGW64 ~/git-skill1 (main)
$ git log --oneline --graph
*   8dd0d4a (HEAD -> main) Resolved merge conflict
| \
|  * ae6a41b (bug-fix) Big fix applied
| * | 9f316d7 (feature-update) Added feature update
| /
* 2e8e9e9 (origin/main) Initial commit with main.java notes.txt
```